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Note: I tried to follow the format of 4 errors for systematicness, so some of the error has itself multiple errors included like you can see in Error 3 of P1.

Paper 1: Kingma & Welling (2013), “Auto-Encoding Variational Bayes.”

Error 1: “The architecture consists of an Encoder network that maps the input x directly to a single point vector z in a lower-dimensional latent space.”

Correction: The encoder (recognition model) does *not* map x to a single point z . Instead, it outputs the *parameters of a distribution* over z specifically, a mean μ and a variance σ^2 , so that $q_\phi(z|x)$ is a full probability distribution (typically a Gaussian). A latent sample is then drawn from this distribution via the reparameterization trick. This is the whole point of a “probabilistic encoder,” as described in Section 2 (page 3) of the paper: “we will refer to the recognition model $q_\phi(z|x)$ as a probabilistic encoder, since given a datapoint x it produces a distribution (e.g. a Gaussian) over the possible values of the code z .”

Error 2: “the authors introduce the Reparameterization Trick, which involves adding uniform noise (sampled from $U[-1, 1]$) to the latent vector”

Correction: The reparameterization trick uses *Gaussian noise*, not uniform noise. Concretely, for a Gaussian $q_\phi(z|x) = \mathcal{N}(\mu, \sigma^2)$, the trick is $z = \mu + \sigma \cdot \epsilon$ where $\epsilon \sim \mathcal{N}(0, 1)$, so the noise is sampled from a standard normal. Pls see Section 2.4 (Eq. in the paragraph after Eq. 6): “let $z \sim p(z|x) = \mathcal{N}(\mu, \sigma^2)$. In this case, a valid reparameterization is $z = \mu + \sigma \epsilon$, where ϵ is an auxiliary noise variable $\epsilon \sim \mathcal{N}(0, 1)$.” While the paper does mention that the inverse-CDF approach uses $\epsilon \sim U(0, 1)$ as one of three listed strategies, the Gaussian example used throughout the paper (and in the VAE itself) draws ϵ from $\mathcal{N}(0, 1)$.

Error 3: “This objective function consists of a discriminative term used to predict the class labels of the data, and a KL Divergence term that acts as a penalty to ensure the approximate posterior $q_\phi(z|x)$ is as dissimilar as possible from the prior $p(z)$.”

Correction: **Two things** are wrong here. First, the ELBO does not contain a “discriminative term for predicting class labels”, it contains a *reconstruction term* $\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]$, which measures how well the decoder can reconstruct the input x from the sampled z . The VAE is a generative model, not a classifier. Second, the KL divergence term encourages the approximate posterior to be as *close* to the prior as possible, not “as dissimilar as possible.” The KL term $-D_{\text{KL}}(q_\phi(z|x)||p(z))$ is *subtracted*, acting as a regularizer that penalizes the approximate posterior for deviating from the prior.

Error 4: “the paper demonstrates through experiments on the Frey Faces dataset that for complex image data, the prior $p(z)$ must be a learned mixture of Gaussians to achieve high-quality reconstruction.”

Correction: The paper uses a simple centered isotropic Gaussian prior $p_\theta(z) = \mathcal{N}(z; 0, I)$ throughout all experiments, including both MNIST and Frey Face datasets and demonstrates good reconstructions with this fixed prior. The paper never requires nor proposes a “learned mixture of Gaussians” prior. As we can see in Section 3 (page 5): “Let the prior over the latent variables be the centered isotropic multivariate Gaussian $p_\theta(z) = \mathcal{N}(z; 0, I)$. Note that in this case, the prior lacks parameters.” Both MNIST and Frey Face experiments use this standard Gaussian prior and achieve high-quality results (pls check Figures 3–5 and Appendix A).

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Paper 2: Ranganath et al. (2014), “Black Box Variational Inference.”

Error 1: “The authors propose an algorithm that only requires the user to provide the analytic gradient of the ELBO with respect to the latent variables z .”

Correction: The whole point of BBVI being “black box” is that the user does *not* need to compute any analytic gradients. The user only needs to provide a function that evaluates the log joint distribution $\log p(x, z)$. The gradient of the ELBO is then estimated via the score function estimator (log-derivative trick) using Monte Carlo samples, so no analytic gradient computation is required from the user. We can see Section 2 (page 2): “From the practitioner’s perspective, this method requires only that he or she write functions to evaluate the model log-likelihood,” and Algorithm 1 text: “the practitioner can compute the log of the joint $p(x, z_s)$.”

Error 2: “control variates are used solely to bias the gradient estimates toward the global optimum, ensuring that the stochastic updates never stray into local minima. These functions must be manually derived for every new model to ensure the gradient variance is exactly zero.”

Correction: This is actually the opposite of what control variates do. Control variates are used to *reduce the variance* of the gradient estimates and they do not introduce bias. The key property of a control variate $h(z)$ is that $\mathbb{E}_q[\nabla \log q(z) \cdot h(z)] = 0$, so subtracting it preserves the expectation (unbiasedness) while reducing variance. Furthermore, the control variates used in BBVI are derived *generically* from the variational distribution itself (using the score function $\nabla_\lambda \log q(z|\lambda)$), so they do *not* require model-specific derivations—this is what makes the method “black box.” Also, they reduce variance substantially but do not make it exactly zero. See the emphasis in Section 3 intro: “We emphasize that these variance reduction methods preserve our goal of black box inference because they do not require computations specific to the model.”

Error 3: “To optimize the parameters, the algorithm uses a fixed, large learning rate.”

Correction: BBVI does not use a fixed learning rate. The paper explicitly uses *adaptive learning rates* via AdaGrad (Duchi et al., 2011), which adjusts the learning rate for each parameter component individually based on the history of gradients. This is critical because different variational parameters may have gradients of very different magnitudes. See Section 1 (page 2): “First, we use adaptive learning rates (Duchi et al., 2011) to set the step size” and the formula for the per-component learning rate $\rho_{t,i}$ based on accumulated squared gradients.

Error 4: “the paper demonstrates BBVI using Stochastic Volatility models, proving the method works even when the model’s log-likelihood $\log p(x, z)$ cannot be evaluated numerically.”

Correction: The paper’s experiments are on *longitudinal medical data* (chronic kidney disease patient data from NY Presbyterian hospital), not on Stochastic Volatility

models. The models tested include a Gamma-Normal time series model and variants. Moreover, BBVI fundamentally *requires* that the log joint $\log p(x, z)$ be evaluable and this is the only thing the user must provide. The algorithm cannot function if the log joint cannot be computed. See Section 5 (page 6): “We use Black Box Variational Inference to quickly construct and evaluate several models on longitudinal medical data,” and Section 2 (page 3): the practitioner must be able to “compute the log of the joint $p(x, z_s)$.”

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Paper 3: Welling & Teh (2011), “Bayesian Learning via Stochastic Gradient Langevin Dynamics.”

Error 1: *“The authors propose a modification to standard SGD where a fixed amount of constant variance noise is added to the gradient updates at every iteration.”*

Correction: The noise injected into the gradient updates does *not* have a fixed, constant variance. Instead, the variance of the injected noise *decreases* over time with the step size: $\eta_t \sim \mathcal{N}(0, \epsilon_t)$, where ϵ_t is the step size that decreases according to a schedule satisfying $\sum \epsilon_t = \infty$ and $\sum \epsilon_t^2 < \infty$. This coupling between step size and noise variance is what balances the gradient signal and the stochastic exploration so that the iterates converge to samples from the posterior. See Eq. (4) and the line below it: “ $\eta_t \sim \mathcal{N}(0, \epsilon_t)$,” where the step sizes decrease towards zero.

Error 2: *“the algorithm eliminates the need for the Metropolis-Hastings (MH) acceptance step by ensuring the step size ϵ_t stays constant throughout training.”*

Correction: It is actually the exact opposite and the algorithm eliminates the MH acceptance step precisely *because* the step size ϵ_t *decreases to zero*. As $\epsilon_t \rightarrow 0$, the discretization error vanishes, the MH rejection rate goes to zero asymptotically, and so the acceptance step can be safely omitted. Keeping ϵ_t constant would actually produce a *biased* chain that requires MH corrections. See Section 3 (page 2): “where the step sizes decrease towards zero at rates satisfying (2). This allows...MH rejection rates that go to zero asymptotically, so that we can simply ignore the MH acceptance steps.”

Error 3: *“The authors prove that as long as the noise is sampled from a Uniform distribution, the distribution of the parameters θ will converge to a single MAP estimate.”*

Correction: Actually two errors here. First, the injected noise is sampled from a *Gaussian* distribution, not a uniform one: $\eta_t \sim \mathcal{N}(0, \epsilon_t)$ (see Eq. 4). This is because the method derives from Langevin dynamics, which is a discretization of a stochastic differential equation with Gaussian (Brownian) noise. Second, the parameters do not converge to a *single MAP estimate* and they converge to *samples from the full posterior distribution* $p(\theta|X)$. Sampling from the posterior (rather than finding a point estimate) is the whole motivation of the method. See Section 3: “we will show that for large t , the updates (4) will approach Langevin dynamics (3), which converges to the posterior distribution.”

Error 4: *“the paper demonstrates the method’s efficiency using a Logistic Regression model on the MNIST dataset.”*

Correction: While the paper does use a Logistic Regression model as one of its experiments, it is applied to the “a9a” dataset (derived from the UCI Adult dataset), not MNIST. The a9a dataset has 32,561 observations and 123 features. The paper’s three experiments are: (i) a simple two-parameter mixture of Gaussians, (ii) logistic regression on the a9a/UCI Adult dataset, and (iii) Independent Components Analysis (ICA). See

Section 5.2: “We applied our stochastic gradient Langevin algorithm to a Bayesian logistic regression model. . . We applied our inference algorithm to the a9a dataset derived by (Lin et al., 2008) from the UCI adult dataset.”

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Paper 4: Lee et al. (2018), “Deep Neural Networks as Gaussian Processes.”

Error 1: “*deep neural networks (DNNs) with a finite number of hidden units per layer are equivalent to Gaussian Processes.*”

Correction: The equivalence holds in the limit of *infinite width*, not for finite-width networks. Specifically, it is when the number of hidden units per layer $N_l \rightarrow \infty$ that the Central Limit Theorem kicks in and the network outputs become Gaussian distributed. Finite-width networks are *not* equivalent to GPs; the paper actually compares finite-width trained networks against the GP and shows they converge as width increases. See Section 2.2: “in the limit of infinite width $N_1 \rightarrow \infty$, $z_i^1(x)$ will be Gaussian distributed,” and the title itself refers to the infinite-width correspondence.

Error 2: “*They prove that to maintain this GP equivalence, one must use a Sigmoid activation function, as ReLU activations cause the covariance to collapse to zero at every layer.*”

Correction: The paper does not restrict itself to Sigmoid activations. In fact, the NNGP construction works for *generic pointwise nonlinearities*, and the experiments use both *ReLU* and *Tanh* activations (not Sigmoid). ReLU activations do *not* cause the covariance to collapse to zero in general, the paper shows that ReLU NNGPs perform well and even gives analytic kernel formulas for ReLU. Both ReLU and Tanh NNGPs achieve competitive performance on MNIST and CIFAR-10, sometimes even outperforming finite-width trained networks. See Section 1.2: “The prescription is valid for generic pointwise nonlinearities,” and Section 3 (experiments): “Nonlinearities were chosen to be either rectified linear units (ReLU) or hyperbolic tangent (Tanh).”

Error 3: “*This equivalence holds because as you add more layers to a network, the Central Limit Theorem (CLT) causes the output to become normally distributed, regardless of the width of the hidden layers.*”

Correction: The CLT applies across *width* (number of units in each layer), not across depth (number of layers). It is the fact that each pre-activation is a sum of many i.i.d. terms (one per unit in the previous layer) that makes the CLT applicable, so we need wide layers, not deep ones. Adding more layers does not by itself invoke the CLT; in fact, very deep networks with finite width can exhibit non-Gaussian behaviour. Pls see Section 2.2: “since $z_i^1(x)$ is a sum of i.i.d terms, it follows from the Central Limit Theorem that in the limit of infinite width $N_1 \rightarrow \infty$, $z_i^1(x)$ will be Gaussian distributed.”

Error 4: “*The paper introduces a recursive method to compute the Neural Tangent Kernel (NTK), which defines the covariance between inputs in a deep GP.*”

Correction: The paper introduces a recursive method to compute the *NNGP kernel* (Neural Network Gaussian Process kernel), not the Neural Tangent Kernel. The NTK is a different object entirely, introduced in a later paper by Jacot et al. (2018). The NNGP kernel is the covariance function of the GP prior that corresponds to an infinite-width

deep network, and it is computed recursively through the layers via Eq. (5) and Eq. (6) in Section 2.3.

Error 5: *“The GP performance is always strictly worse than the corresponding finite-width trained neural network ... the GP derived from a 10-layer CNN outperforms a standard SVM.”*

Correction: The paper does not claim that Neural Network Gaussian Processes (NNGPs) are inferior to finite-width networks. On the contrary, (section 4) experiments on datasets like MNIST and CIFAR-10 demonstrate that NNGPs often match or even outperform finite-width networks (especially those trained without advanced optimization techniques). The paper presents the NNGP as a strong, competitive baseline.”

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Paper 5: Kirsch et al. (2019), “BatchBALD: Efficient and Diverse Batch Acquisition for Active Learning.”

Error 1: “standard BALD is inefficient for batch acquisition because it underestimates the uncertainty of individual points, leading the model to ignore outliers.”

Correction: The problem with naively using BALD for batches is not that it underestimates uncertainty. Rather, BALD scores each point *independently*, so when you take the top- b points, you end up acquiring points that are individually informative but *highly redundant* with one another. The batch may consist of near-duplicate data points that all provide roughly the same information about the model parameters. See Section 1 and Figure 1: “If a dataset were to contain many (near) replicas for each data point, then BALD would select all replicas of a single informative data point at the expense of other informative data points, wasting data efficiency.”

Error 2: “BatchBALD maximizes the sum of the individual mutual informations of all points in a candidate batch. This objective ensures that the points in the batch are as similar to each other as possible to provide a ‘consensus’ update.”

Correction: Maximising the *sum of individual mutual informations* is exactly what naive top- k BALD does, and that is the baseline BatchBALD is designed to improve upon. BatchBALD instead maximises the *joint mutual information* $I(y_1, y_2, \dots, y_b; \omega | x_1, \dots, x_b)$ between all the labels jointly and the model parameters. This joint mutual information accounts for redundancy between points and actively encourages *diversity* in the batch, not similarity. See Section 3.1 and Eq. (4) where BatchBALD is defined as the joint mutual information, contrasted with Eq. (3) which is the sum of individual scores used by naive BALD.

Error 3: “The paper proves that this objective is linearly scalable, meaning that calculating a batch of size 1000 is no more computationally expensive than calculating a batch of size 1.”

Correction: The paper shows that the greedy algorithm runs in *linear time* in the acquisition batch size b (i.e., $O(b)$), meaning cost grows linearly with b and not that it is constant. A batch of size 1000 is roughly 1000 times more expensive than a batch of size 1, but this is still efficient compared to the exponential cost of exhaustive search over all subsets. The algorithm is also a $1 - 1/e$ approximation to the optimal by virtue of the objective’s submodularity. See Section 1/Abstract: “BatchBALD is a greedy linear-time $1 - 1/e$ approximation,” and Section 3.2 on the greedy algorithm.

Error 4: “the authors demonstrate that BatchBALD is particularly effective for Support Vector Machines (SVMs) because they lack the ‘overconfidence’ found in Neural Networks.”

Correction: The paper uses *Bayesian Neural Networks* (BNNs) with MC Dropout as the model, not SVMs. SVMs are never used or even discussed as a candidate model in

this paper. The method requires a Bayesian model that provides a posterior distribution over parameters so that mutual information can be estimated, SVMs do not naturally provide this. Please see Section 2.3: “In this paper we focus on BNNs as our Bayesian model because they scale well to high dimensional inputs, such as images,” using MC dropout for approximate inference.

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Paper 6: Fort, Hu & Lakshminarayanan (2019), “Deep Ensembles: A Loss Landscape Perspective.”

Error 1: *“This paper provides a theoretical justification for deep ensembles by proving that they are equivalent to exact Bayesian inference in neural networks.”*

Correction: The paper does *not* prove any equivalence to exact Bayesian inference. In fact, the paper explicitly notes that deep ensembles are *not* performing exact Bayesian inference, each ensemble member converges to a MAP estimate (a single mode), and the ensemble combines these distinct modes. The paper’s contribution is to investigate *why* ensembles work well from a loss landscape perspective, hypothesising that independently initialised models explore *diverse modes in function space*, which gives them better predictive diversity than variational Bayesian methods that stay within a single mode. See the Abstract: “One possible explanation for this gap between theory and practice is that popular scalable variational Bayesian methods tend to focus on a single mode, whereas deep ensembles tend to explore diverse modes in function space.”

Error 2: *“models obtained along a single SGD training trajectory explore multiple distinct modes of the posterior. In contrast, models initialized independently tend to converge to the same mode due to the strong convexity of the loss function.”*

Correction: This is exactly backwards. The paper shows that models along a *single* SGD trajectory (checkpoints along training, or samples from a subspace around the optimum) cluster *within a single mode* in function space and produce very similar predictions. Conversely, models *independently initialised* with different random seeds converge to *different, distinct modes* in function space. Also, the loss function of deep networks is fundamentally *non-convex* (the paper mentions this explicitly), which is precisely why multiple modes exist. See Section 4.1 and Figure 2: “functions sampled along a single training trajectory or subspace thereof cluster within a single mode predictions-wise,” and Section 1: “which could lead to M different solutions, if the loss is non-convex.”

Error 3: *“The authors introduce a diversity–accuracy tradeoff and conclude that increasing diversity always harms predictive accuracy, making large ensembles undesirable.”*

Correction: The paper introduces the diversity–accuracy *plane* to visualise how different methods trade off between the two, but it does *not* conclude that diversity always harms accuracy. Rather, the paper shows that deep ensembles achieve a *favourable* trade-off, with high diversity *and* high accuracy simultaneously. The core finding is that the decorrelation power of random initialisation is unmatched by subspace sampling methods. Deep ensembles are shown to outperform alternatives, especially under dataset shift, making large ensembles *desirable*. See Section 4.3 and Figure 6: “Deep ensembles consistently achieve a better diversity vs accuracy trade-off than the four different subspace sampling methods.”

Error 4: “the experiments demonstrate that subspace sampling methods such as SWAG consistently outperform deep ensembles on out-of-distribution generalization tasks.”

Correction: The experimental results show the opposite. Deep ensembles *outperform* subspace sampling methods (including SWAG-like diagonal/low-rank Gaussian, dropout, and random subspace methods) on out-of-distribution tasks. The paper attributes this to the fact that ensembles sample from *different modes* in function space, providing greater functional diversity than subspace methods that remain in the vicinity of a single mode. See Section 4.4 and Figure 8–9: ensembles achieve better accuracy and Brier scores, and the diversity gap between ensembles and subspace methods *widens* under dataset shift (CIFAR-10-C).

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Paper 7: Kucukelbir et al. (2017), “Automatic Differentiation Variational Inference.”

Error 1: *“The method relies on a reparameterization of the variational distribution that eliminates the need for stochastic gradient estimates, allowing exact gradients of the ELBO to be computed via automatic differentiation.”*

Correction: ADVI does *not* compute exact gradients of the ELBO. The ELBO still involves an intractable expectation, and so ADVI uses *Monte Carlo* estimates of the gradient, the reparameterization trick converts the expectation into one over a standard Gaussian, from which samples are drawn to form a noisy (but unbiased) gradient estimate. Automatic differentiation is then used to compute gradients of the integrand, but the overall gradient estimate is still stochastic. We can clearly see this in Section 2.5: “We compute the elbo for any model using Monte Carlo (MC) integration, which only requires being able to sample from the variational distribution,” and Step 4 in Section 1: “advi uses noisy gradients to optimize the variational distribution.”

Error 2: *“In practice, ADVI is implemented using a mean-field Gaussian approximation, as the paper shows that full-rank covariance structures offer negligible practical benefits.”*

Correction: Firstly, ADVI offers *both* mean-field and full-rank Gaussian approximations, and the paper explicitly shows that full-rank ADVI provides *substantial* benefits when posterior correlations are important. For instance, in the stochastic volatility model experiment, mean-field ADVI fails to capture the correlated structure of the posterior, underestimating the volatility in regions where the log-volatility deviates from its mean. Full-rank ADVI closely matches the MCMC sampling results in both marginal variances and posterior correlations. See Section 3.1 (Figure 4): the full-rank approximation accurately recovers the posterior, and the recommendation in Section 3.1: “Scientists interested in posterior variances and covariances should use the full-rank approximation.”

Error 3: *“A key advantage of ADVI is that it avoids the need for model-specific derivations by assuming that all latent variables are independent in the variational family, which simplifies inference without significantly affecting posterior correlations.”*

Correction: While the mean-field variant does assume independence across the latent variables, the paper *does not* claim this assumption is harmless for posterior correlations and in fact, it explicitly shows the opposite. Mean-field ADVI can miss important posterior correlations, and this can lead to poor variance estimates. The real key advantage of ADVI is the *automatic transformation* of constrained latent variables to an unconstrained space (Step 1), combined with automatic differentiation of the log joint, which together make the method model-agnostic without requiring manual derivations. See Section 2 (Steps 1–4) and Section 3.1 where the “correlation” structure is only captured by the full-rank approximation.

Error 4: *“the empirical evaluation demonstrates that ADVI consistently matches or*

exceeds the accuracy of Hamiltonian Monte Carlo across all tested models while being significantly faster.”

Correction: ADVI is indeed faster than HMC/NUTS (often by an order of magnitude), but it does *not* consistently match or exceed their accuracy across all models. The paper shows that ADVI is an *approximation*—mean-field ADVI can underestimate posterior variances, and even full-rank ADVI can deviate from the MCMC ground truth in some models. For example, in the logistic regression experiment (Figure 5), mean-field ADVI underestimates marginal posterior variances on most coefficients. The paper positions ADVI as a fast, practical alternative that trades off some accuracy for speed, not as a method that dominates MCMC in accuracy. We can clearly see this in Section 4 (Figures 5–7 and 11).

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Paper 8: Daxberger et al. (2021), “Laplace Redux – Effortless Bayesian Deep Learning.”

Error 1: *“the authors show that different approximations (e.g., diagonal, KFAC, and full-matrix forms) lead to similar predictive performance when applied to modern neural networks.”*

Correction: The paper tells that different curvature approximations lead to *meaningfully different* performance, not similar performance. In particular, *diagonal* approximations are found to perform *significantly worse* than KFAC (Kronecker-factored approximate curvature), and this difference matters in practice. The paper explicitly recommends against using diagonal approximations. See Section 4 (Recommendations): “Diagonal approximations perform significantly worse than KFAC and are therefore not suggested..”

Error 2: *“A key contribution of the paper is demonstrating that applying LA to only the final layer provides a computationally efficient approximation that closely matches full-network Laplace across a wide range of tasks.”*

Correction: The paper actually finds that last-layer LA typically *outperforms* all-layer LA (i.e., performs *better*, not just “closely matches”), because the all-layer Laplace tends to *underfit* due to the crudeness of the Gaussian approximation applied to the full high-dimensional weight space. The last-layer approach avoids this problem and also has minimal computational overhead. See Section 4 (Recommendations): “last layer instead of all weights typically yields better performance due to less underfitting..”

Error 3: *“The authors further argue that marginal likelihood optimization for tuning prior precision is largely unnecessary in practice, as predictive performance is relatively insensitive to this hyperparameter.”*

Correction: The paper actually argues the opposite and it highlights marginal likelihood optimisation as an *important* and *useful* tool for hyperparameter tuning. In fact, marginal likelihood optimisation is presented as a key advantage of the Laplace approximation over competing methods (which typically require held-out validation data for tuning). The paper shows that online marginal likelihood training can lead to improved performance and even enable better model selection. See Section 3 (Hyperparameter Tuning): “marginal likelihood maximization (a.k.a. empirical Bayes)... requires no validation data,” and Immer et al. (2021) results cited in Section 4.4 about online marginal likelihood tuning.

Error 4: *“their performance is largely invariant to the specific predictive approximation used.”*

Correction: The choice of predictive approximation *does* matter in practice. The paper explicitly compares different predictive approximations like MC integration, the probit approximation, and the Linearised Laplace (bridge) approximation and finds that

they differ in quality and cost. For the last-layer LA, the *probit approximation* provides the best trade-off, while the bridge approximation “typically provides inferior results.” For all-layer LA, MC integration is preferred since the probit approximation is not applicable. See Section 4 (Recommendations): “the probit provides the best approximation to the predictive for the last-layer LA. MC integration is needed for all-layer predictions. The bridge has the same cost as the probit approximation but typically provides inferior results in our experiments.”